

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

Kinetic Theory

LECTURE-02 PHYSICS

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Topics To Be Covered

- 1** Revision
- 2** Gaseous Law
- 3** Ideal gas Equation
- 4** Root Mean Square Velocity of Gas molecule
- 5** ..

Recap from 1st Lecture

1 KTG

1
2
3
4
5
6
7
8
9
10



$$P = \frac{1}{3} \rho v^2$$

$$P = \frac{2}{3} \frac{K.E}{V}$$

Pressure and K.E
Relationship

$$K.E = \frac{3}{2} nRT$$

K.E and Temperature

Mean Free path

Gaseous law

⇒ The laws which are followed by **gas**

⇒ There are **4** gaseous laws

(a) Boyle's law

(b) Gay-Lussac's law

(c) Charles's law

(d) Avogadro's law

① Boyle's law (at const. temp)

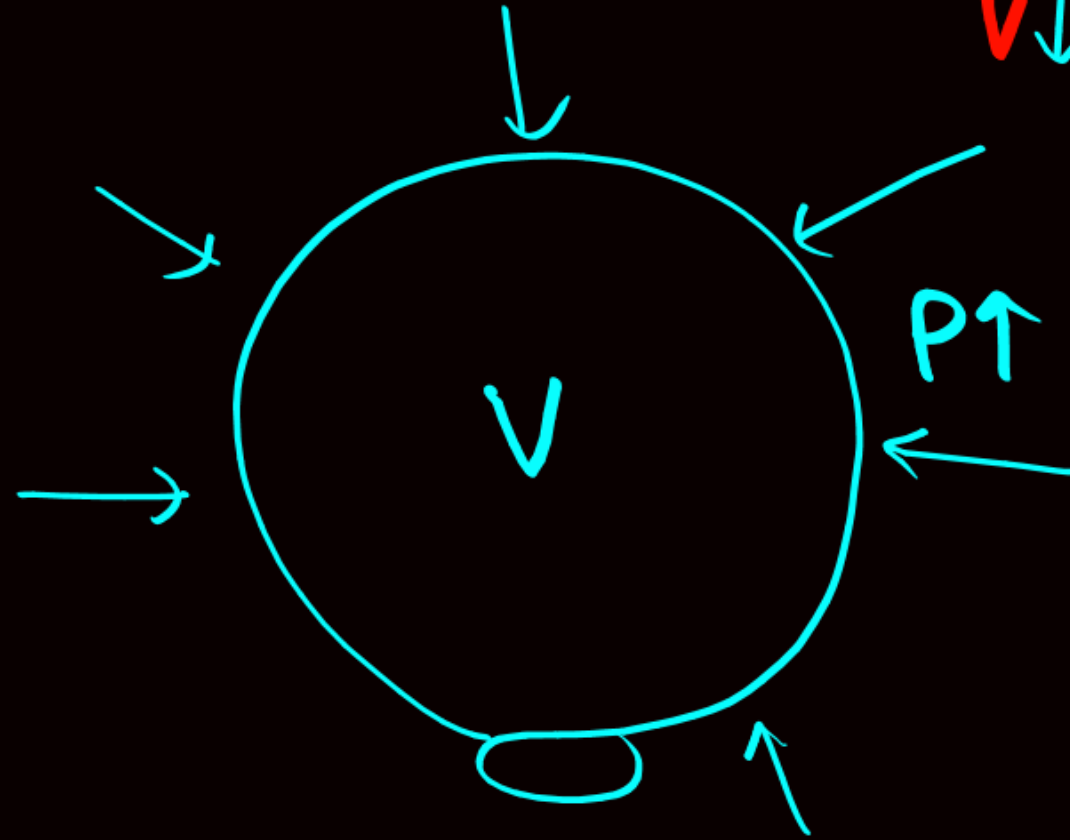
The pressure is inversely proportional to the volume

Pressure $\propto \frac{1}{\text{Volume}}$

$$P \propto \frac{1}{V}$$

$$P = k \times \frac{1}{V}$$

$$\uparrow P = \frac{k}{V \downarrow}$$

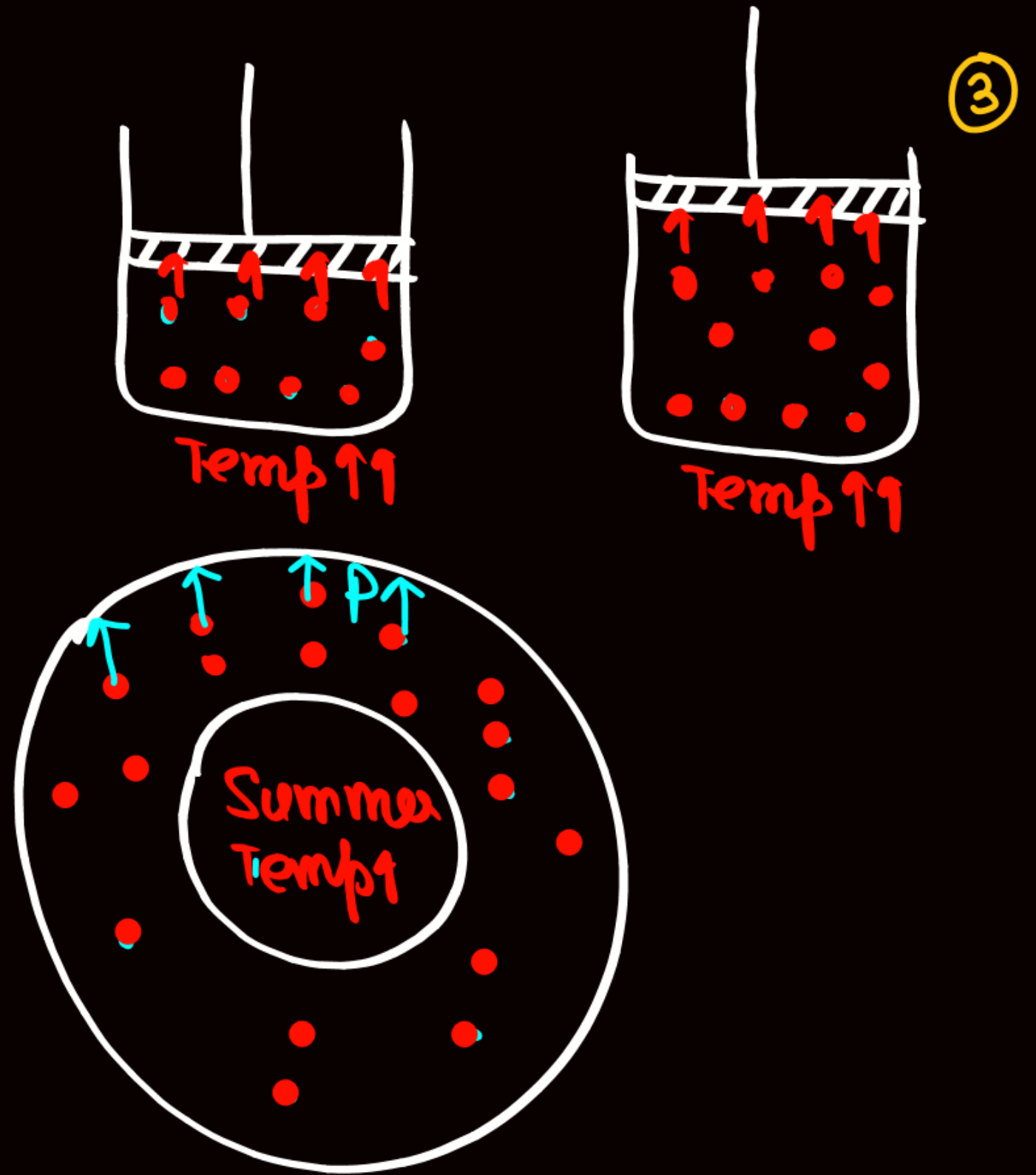


②

(2) Gay-Lussac law

- # It is the relationship b/w pressure and temperature (PT)
- # The pressure is directly proportional to the temperature

$$P \propto T$$
$$\uparrow P = k T \uparrow$$



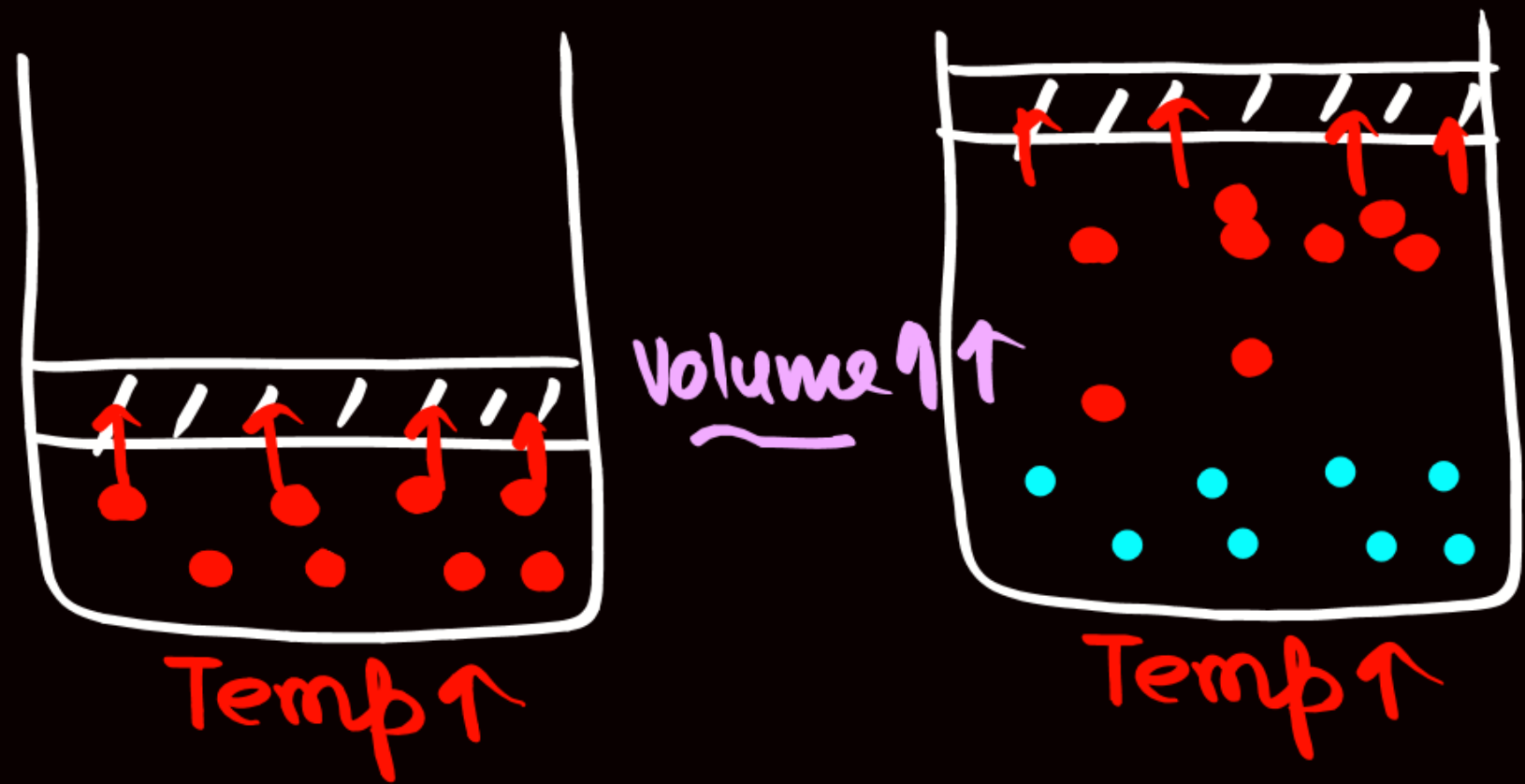
③ Charles law

It is the Relationship
b/w Volume and
temperature

Volume is directly
Proportional to temp

$$V \propto T$$

$$V = KT$$



④

(4) Avagadro's law

It is the relationship

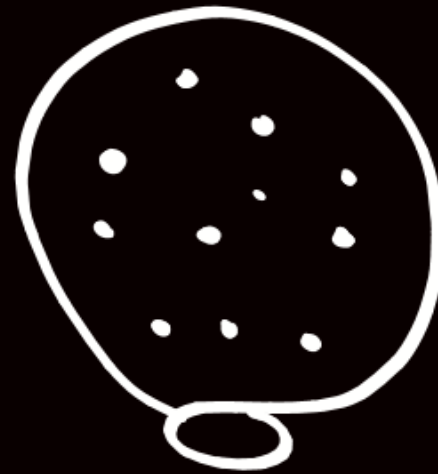
blw no. of moles of gas

and volume

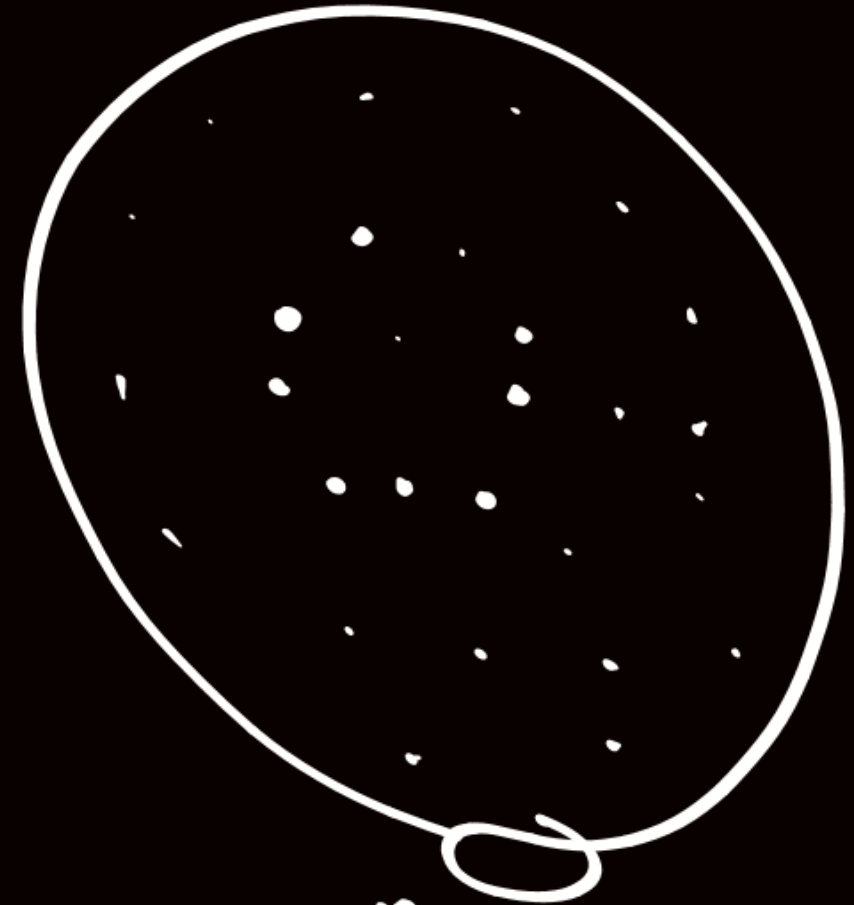
$$V \propto n$$

$$V = k r$$

↳ constant



$n \downarrow$
 $V \downarrow$



$n \uparrow$
 $V \uparrow$

(5)

Ideal gas equation

Combination of 4 gaseous law

$$P \propto \frac{1}{V} \text{ --- (I) Boyle's law} \checkmark$$

$$P \propto T \text{ --- (II) G-L-law} \checkmark$$

$$\underline{V} \propto \underline{T} \text{ --- III Charles law} \checkmark$$

$$V \propto n \text{ --- IV Avagadro's law}$$

From above knowledge we
design the Ideal gas equation

- $P \propto \frac{T}{V}$

$$\underline{PV} \propto nT$$

$$PV = RnT$$

$$\boxed{PV = nRT}$$

$R \rightarrow$ universal gas constant

$$R = 8.14 \frac{\text{Joule}}{\text{mole K}}$$

⑥

Average / Mean velocity of gas molecule

⑦

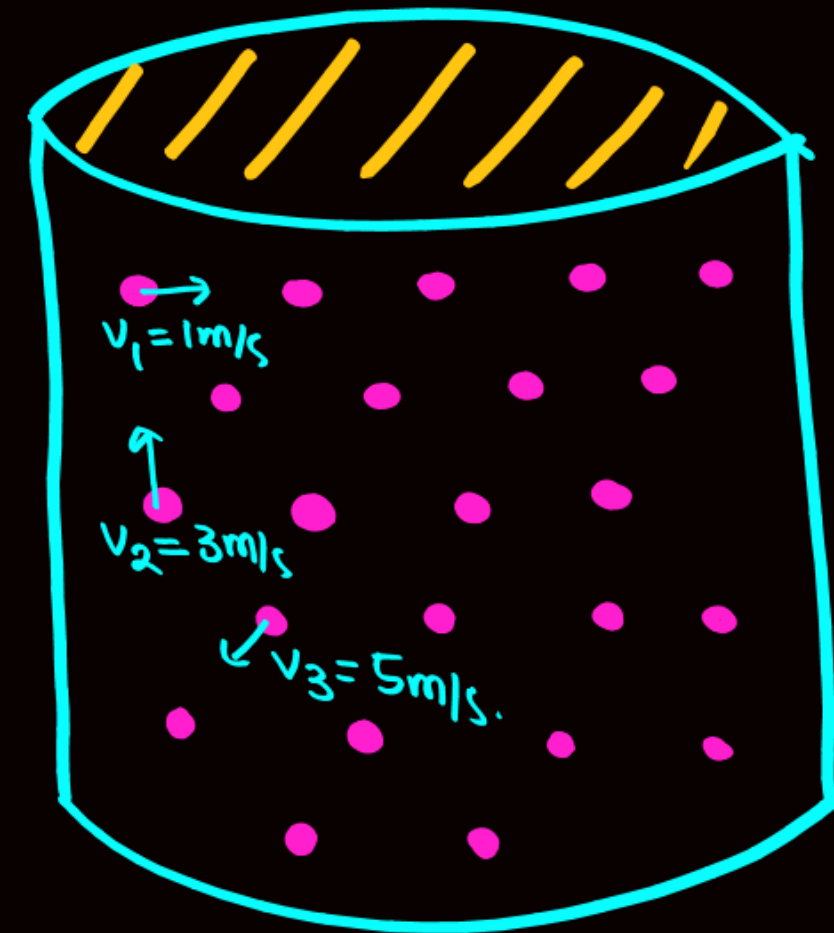
Every gas molecule moves with different velocity

$v_1, v_2, v_3, v_4, v_5, \dots, v_n$

$$V_{\text{average / mean}} = \frac{\text{Sum of All observation}}{\text{no. of observation}}$$

$$V_{\text{average}} = \frac{v_1 + v_2 + v_3 + v_4 + \dots + v_n}{n} = \sqrt{\frac{8RT}{\pi M}}$$

V_{mean}



8

$$\downarrow \uparrow V_a = \sqrt{\frac{8R \textcircled{T} \uparrow}{\pi \textcircled{m} \uparrow}}$$

no derivation

molecular mass
of gas $\uparrow$

numerical
Re-Neet 2026

Root Mean Square Velocity

V_{rms}

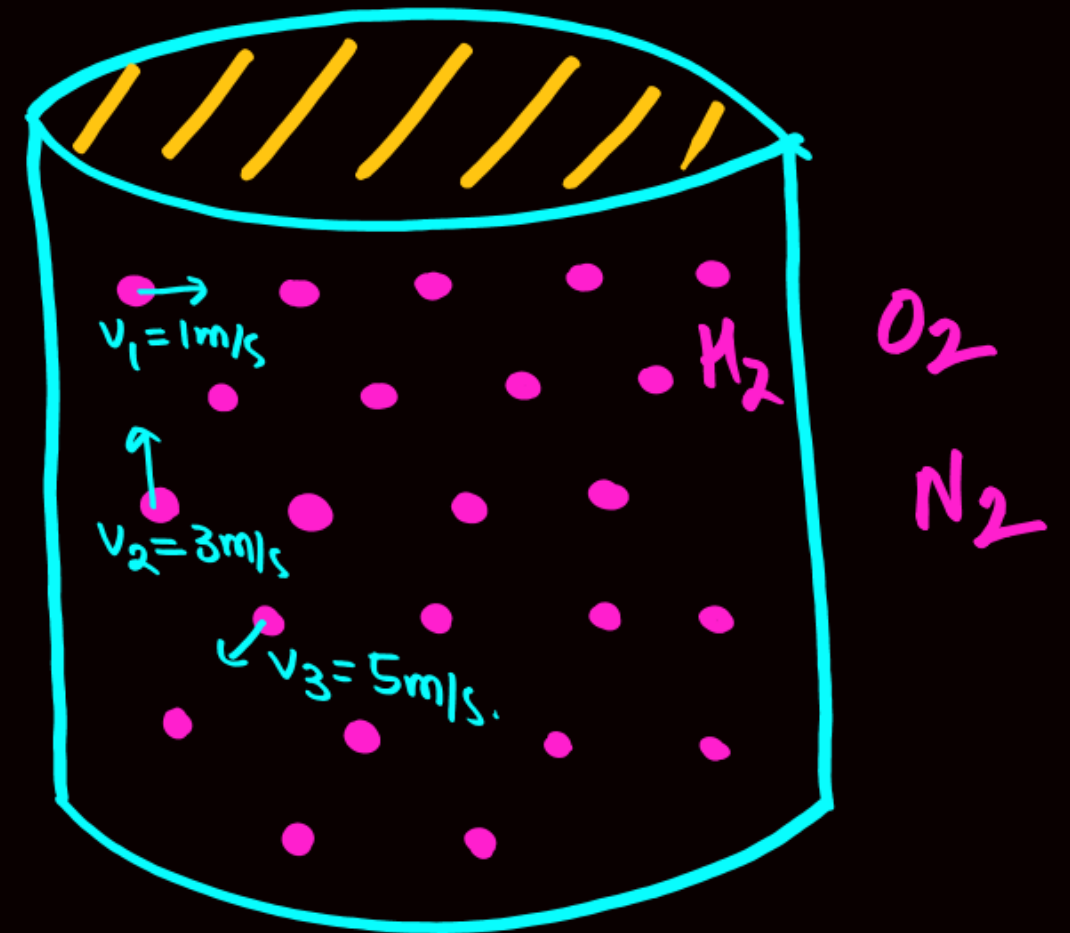
⑨

Every gas molecule moves with different velocity

$v_1, v_2, v_3, v_4, v_5, \dots, v_n$

$$V_{rms} = \sqrt{\frac{v_1^2 + v_2^2 + v_3^2 + \dots + v_n^2}{n}}$$

$$V_{rms} = \sqrt{\frac{3RT}{m}}$$



$\downarrow \uparrow \uparrow V_{rms}$

=

$$\sqrt{\frac{3RT}{m}}$$

Temperature (10)

molecular mass of gas

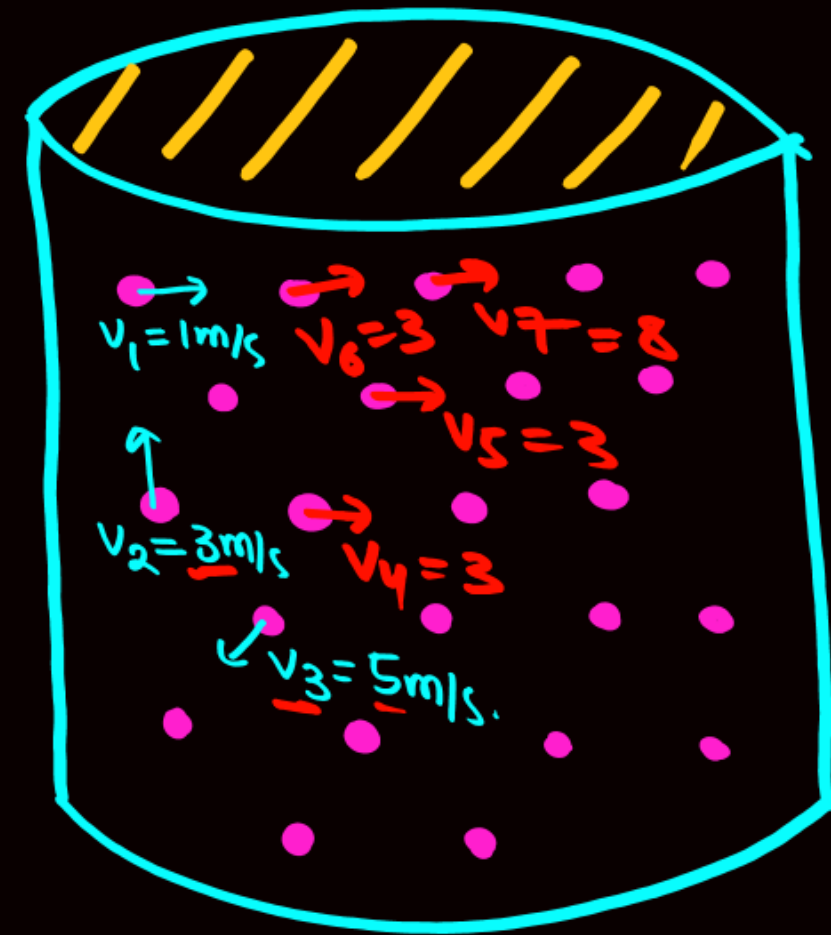
③ Most Probable velocity

Every gas molecule moves with different velocity

$\frac{v_1}{1}, \frac{v_2}{3}, \frac{v_3}{5}, \frac{v_4}{3}, \frac{v_5}{3}, \frac{v_6}{3}, \frac{v_7}{8}, \frac{v_n}{8}$

The velocity with which the maximum number of molecule moves.

$$\# V_{mpv} = \sqrt{\frac{2RT}{m}}$$



②

$$V_{mpv} = \sqrt{\frac{2RT}{m}}$$

Temperature
molecular mass.

Mole Concept

Society

1 Dazon = 12

$$2D = 2 \times 10$$
$$= 2 \times 12 = 24$$

$$3D = 3 \times 10 = 3 \times 12 = 36$$

$$4D = 4 \times 1D = 4 \times 12 = 48$$

$$50 = 5 \times 10 = 5 \times 12 = 60$$

chemistry

$$1 \text{ mole} = 6.023 \times 10^{23}$$

[illegible]

$$1 \text{ mole} = 6 \times 10^{23}$$

$$= \begin{matrix} & 60000000000000000 \\ & 00\ 0\ 0\ 0\ 0\end{matrix}$$

$$\text{no. of mole} = \frac{\text{Given weight}}{\text{Molecular weight}}$$

$$\text{no. of mole} = \frac{\text{Given Volume}}{22.4 \text{ lt}}$$

$$\text{no. of moles} = \frac{\text{Given number}}{6 \times 10^{23}}$$

H/w

$$V_{a/m} = \sqrt{\frac{8}{\pi} \frac{RT}{M}}$$

$$V_{RMS} = \sqrt{3 \frac{RT}{M}}$$

$$V_{MPV} = \sqrt{2 \frac{RT}{M}}$$

Neet - 2019 ✓

$$\frac{V_a}{V_{RMS}} = \frac{V_{MPV}}{V_{RMS}} = ?$$

Mehnat Karo Mehnat karo
Mehnat karne se ho jata hai
Insallah apke 498 No. Aayege
Bas Yhi motivation apne dil me
rkho bs



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Thank
You

